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Development and Validation of a Robust High-Speed Mud Pulse Telemetry System for Versatile Drilling Applications

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Abstract

The industry trend to drill wells faster and with greater precision in increasingly challenging formations results in a growing need for real-time measurement-while-drilling (MWD) and logging-while-drilling (LWD) data to manage and control the drilling process optimally. Mud pulse telemetry systems are expected to offer higher bandwidth to satisfy this ever-increasing demand for data while becoming more reliable (see [Kimberlite 2025](#)), especially temperature and lost circulation material (LCM) tolerant. This paper presents a next generation mud pulse telemetry system that satisfies these traditionally mutually exclusive requirements. The telemetry system is built around a new reciprocating mud pulser, whose concept was developed to combine LCM tolerance and high temperature capability with a high system dynamic to generate high frequency pressure signals exceeding 30 Hz. For the system design, the telemetry system was decomposed into its main subsystems pulser, signal modulation and receiver. The subsystems were conceptualized, iteratively designed and simulated considering the dependencies between subsystems to optimally use the capabilities of the new pulser concept. Various laboratory tests were conducted to verify design decisions on subsystem level, followed by field tests to validate overall system performance.

The modeled flow regime and pressure buildup of the reciprocating mud pulse valve agreed very well with measurements taken in laboratory tests. Consequently, plugging tests with lost circulation material also showed positive results, even with high frequency signals. An autoclave test confirmed the high temperature capability. Drilling operations at a test rig were conducted to verify the implementation of the newly developed signal modulation scheme with its corresponding decoding algorithms and MWD surface software. Afterwards, system deployments in performance drilling applications in > 20 unique wells with drilled footage exceeding 250,000 ft and more than 1,700 circ hours validated ruggedness of the pulser system. A physical data rate of up to 18 bit/sec could be achieved in an environment where data rates of up to 3 bit/sec are the norm.

In summary, it could be confirmed that high-frequency mud pulse systems can be highly LCM tolerant and high temperature capable. This makes the presented system operationally a lot more versatile than systems established in the industry. Wells can be drilled with high rate-of-penetration (ROP) without compromising LWD log density. Moreover, wellbore strengthening operations can be conducted without an increased risk of non-productive time (NPT) to trip for replacement of a plugged MWD pulser. A holistic approach to mud pulse telemetry system design delivered a versatile and highly reliable high-speed mud

pulse telemetry system. Making high resolution real-time data available in all kinds of drilling applications, e.g. brownfield developments, enables increased drilling performance and accurate wellbore placement using advanced LWD measurements, and at the same time reduces the risk of exceeding well authorization for expenditure (AFE).

Introduction

In the ever-evolving landscape of oil and gas exploration and production, directional-drilling and logging-while-drilling (LWD) are established, indispensable technologies for efficient well construction. These technologies allow operators to access difficult-to-reach reservoirs, optimize wellbore placement, and gather critical downhole data in real-time for fast decision making. As the industry pushes into more challenging frontiers, such as deeper waters, unconventional resources, and high-pressure, high-temperature (HPHT) environments, the demands on drilling and measurement technologies are steadily increasing. Navigating complex reservoirs requires accurate, high-resolution MWD & LWD data transmitted by a telemetry system to understand the reservoir behavior, identify sweet spots, steer the wellbore in the target zone and avoid hazards. By continuously monitoring formation properties, operators can make immediate corrections to ensure optimal well placement. This maximizes hydrocarbon recovery while minimizing the risk of drilling operations, which can compromise well productivity. Moreover, a telemetry system that offers faster data transmission rates enables enhanced drilling efficiency. It is a requirement for drilling systems automation to enable automated directional drilling ([Hansen et al 2020](#)) and reservoir navigation ([Wessling et al 2023](#)), and other new methodologies such as continuous rotational surveying ([Anda et al 2024](#)). Real-time access to downhole data allows operators to adjust drilling parameters, such as weight on bit, rotary speed, and mud flow, to optimize the rate of penetration (ROP)—see, for example, [Behounek et al 2023](#). This leads to faster drilling time to the well target and a reduction of non-productive time (NPT). In an industry where operational efficiency is critical, these benefits are invaluable. Established telemetry systems, however, may struggle to provide the necessary breadth of MWD/LWD data at sufficient log density, e.g. in extended-reach drilling (ERD) or horizontal wells where signal attenuation is a concern.

From a drilling point of view, an additional operational challenge is the growing number of wells with increased risk of fluid losses into the formation. This issue is common in fractured or highly permeable zones and can lead to reduced wellbore stability, increased risk of stuck pipe, and potential blowouts. To mitigate fluid losses and strengthen the wellbore, operators often pump lost circulation material (LCM) into the well, either as pill/sweep or continuously. LCMs are fibrous, flaky or granular materials added to the mud system to bridge formation fractures, fill voids and seal off a zone of fluid losses. LCM can lead to an increased risk of pulser blockage resulting in a disruption of real-time data transmission, which in most cases requires a trip to replace the plugged MWD pulser.

These operational challenges presented above push the limits of equipment reliability and transmission rate. Consequently, there is an urgent need for reliable high speed telemetry systems (see [Kimberlite 2025](#)) that can deliver high-resolution data in real-time even under the most adverse conditions like high-temperature environments and wellbore strengthening operations.

In the remainder of this paper, we will present a next generation pulser/telemetry system designed to provide high resolution real-time data, combined with improved LCM tolerance and superior performance in extreme environmental conditions. This innovative system will empower operators to make quicker, more informed decisions, ultimately leading to safer, and more cost-effective drilling operations to deliver wells within AFE.

Mud Pulse Telemetry System Overview

Essentially mud pulse telemetry systems are digital communication systems adapted for an extremely harsh operating environment. Their system components can be mapped onto the generic structure of a digital

communication system (Figure 1). Within a drilling BHA various data sources (MWD / LWD data streams) are competing for the scarce resource of mud pulse telemetry bandwidth. Measurement specific lossless or lossy "data compression" algorithms are implemented to minimize bandwidth needs. Lossless algorithms eliminate redundancy in the measurement data, while lossy algorithms remove information that is less important / not relevant for real-time decision making. Typically, compression algorithms are measurement specific. After compression the data streams from the various data sources are serially combined into a single bit stream. Within the industry the amount of uncompressed data transmitted within a given period of time, i.e. the transmission rate of the uncompressed data, is referred to as "effective data rate". In contrast "net data rate" or "throughput" is used when referring to the transmission rate of the compressed data stream. Last but not least, "physical data rate" is used when referring to the actual transmission rate in the channel including all telemetry overhead such as channel coding, synchronization signals etc. All three data rates describe specific aspects of the telemetry system. The "effective data rate" is by far the highest of the three. In some cases, e.g. highly compressed LWD images, the instantaneous "effective data rate" of a mud pulse telemetry system can well exceed 100 bit/sec (see Mazerov, K. 2013).

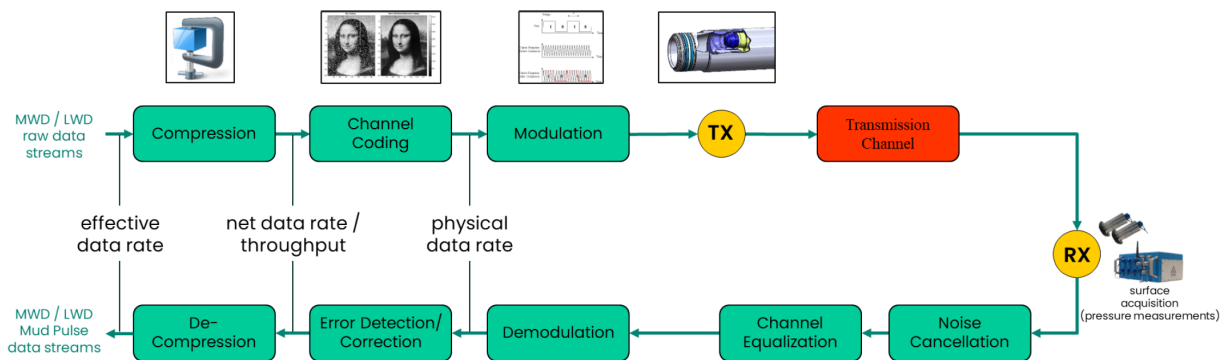


Figure 1—Mud Pulse Telemetry System mapped onto the main components of a digital communication system

Since the mud pulse telemetry channels are prone to transmission errors, "channel coding" is used to add redundant bits to the compressed and serialized bit stream allowing the mud pulse decoder on surface to detect and, in some cases, even correct transmission errors. "Modulation" transforms the protected bit stream into a pressure waveform, which drives the transmitter "TX" (i.e. the mud pulser). In the "transmission channel" the pressure wave signals from the mud pulser suffer from interfering pressure fluctuations, signal reflections and attenuation. On the receiver side, on surface, pressure sensors serve as receiver antennas RX. "Noise cancellation" and "channel equalization" algorithms are used to reconstruct the waveform transmitted by the pulser. The "demodulator" recreates the protected bit stream. The data transmission capability of the signal modulation scheme, the mud pulser, the transmission channel (i.e. the mud system), and the receiver system (i.e. pressure sensors, algorithms for signal reconstruction and demodulation) is described by the "physical data rate". It heavily depends on the signal attenuation and noise level of the mud channel as well as on the design of the pulser and modulation system to optimally exploit the mud channel capabilities. "Physical data rates" are within a range of 1 bit/sec to 40 bit/sec. Since the "physical data rate" includes all redundancy introduced by channel coding, it is slightly higher than the "net data rate".

In this paper we focus on the design and verification of a novel mud pulser, the signal modulation system optimally using the pulser capabilities and the associated signal reconstruction and demodulation algorithms.

Robust High-Speed Mud Pulser

The industry is using different technologies and devices to generate pressure fluctuations in the mud stream for telemetry. In general, fluid pressure variations are generated by moving a valve with respect to a

restriction in an oscillating manner, creating pressure differentials that are transmitted through the borehole fluid to a receiver at surface. Valves for this purpose may be rotating (see for example, [Hutin et al 2001](#)) or oscillating disc valves (see for example, [Klotz et al 2008](#)) or reciprocating plunger valves (see for example, [Hahn et al 2008](#)). Many service companies rely on hydraulically assisted pulser devices that use pilot valves to actuate the signal pressure generating valve (see for example, [Spinnler 1976](#)). Such devices have common constraints, which are a limited actuation frequency, no controlled valve piston positioning and they are prone to plugging of the required bypass. While rotating disc valves are perceived to be superior as high-speed telemetry devices, direct driven reciprocating plunger valves (not pilot valve assisted) are known to offer much higher robustness against plugging from debris or any kind of (coarse) LCM. Since such reciprocating valves can be positioned precisely according to the current downhole condition of flow rate and fluid density, signal pressure can be adjusted as needed to match telemetry requirements. However, plunger valves work against the mud stream while creating pressure signals and thus demand high actuation forces to position the valve close to the restrictor ([Figure 2](#)). A typical pressure plot, generated by a reciprocating valve, is depicted in [Figure 3](#) (left). The position on the X-axis represents the distance between plunger and restrictor. The pressure shows a logarithmic decrease with increasing distance between restrictor and valve plunger. The required valve force to hold the plunger in position shows an analogue behavior ([Figure 3](#), right)—see [Hahn et al 2008](#).

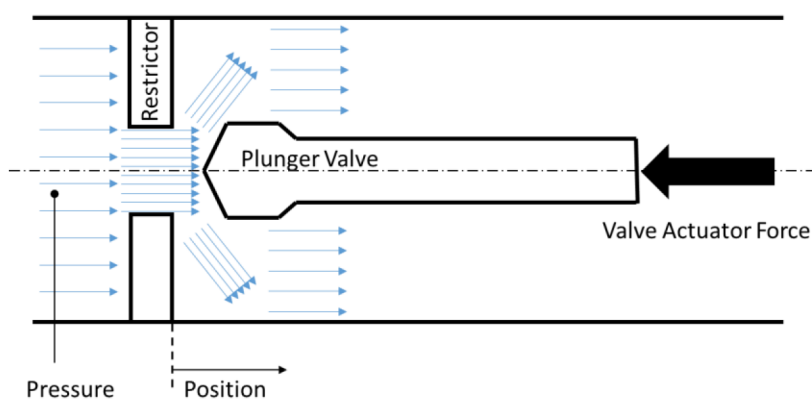


Figure 2—Working principle of plunger valve with reciprocating plunger movement relative to a restrictor to create a pressure signal

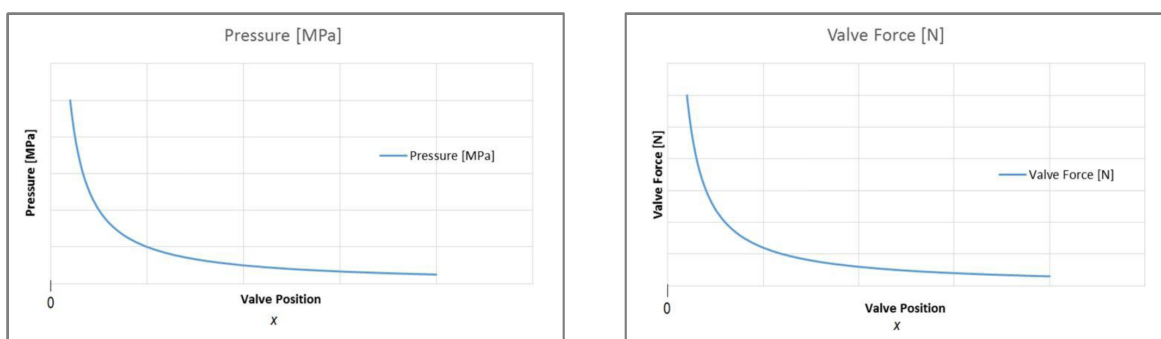


Figure 3—Distance between plunger and restrictor (Valve Position) influencing the generated pressure (left) and required actuating force (right)

In an exemplary application, the force to create a signal pressure of e. g. 2 MPa can reach several kN, depending on the geometry of the valve head and the restrictor. Such a high force makes it extremely difficult to design a direct driven reciprocating valve for high data rate telemetry with signal frequencies well beyond 10 Hz actuation. In a high-speed telemetry application such a direct driven reciprocating valve

would require more than 1 kW of mechanical power to be actuated. To overcome this limitation of direct driven reciprocating valves, a new pressure pulse generator was designed, built and tested.

Force Balanced Pulser

Introducing a secondary plunger piston in hydraulic communication to the high-pressure side of the valve arrangement (upstream side of the restrictor) and a mechanical connection, such as a rocker lever between the reciprocating valve and the secondary plunger piston, allows for balancing of hydraulic valve forces (Figure 4). Whenever the plunger valve closes and the pressure rises, the secondary plunger piston picks up this same pressure and thus creates a piston force upon this current in-situ pressure. Joining valve and secondary piston plungers by a suitable mechanical connection and selecting lever arms and piston surfaces carefully, forces can be efficiently balanced (Figure 5)—see Peters, 2024.

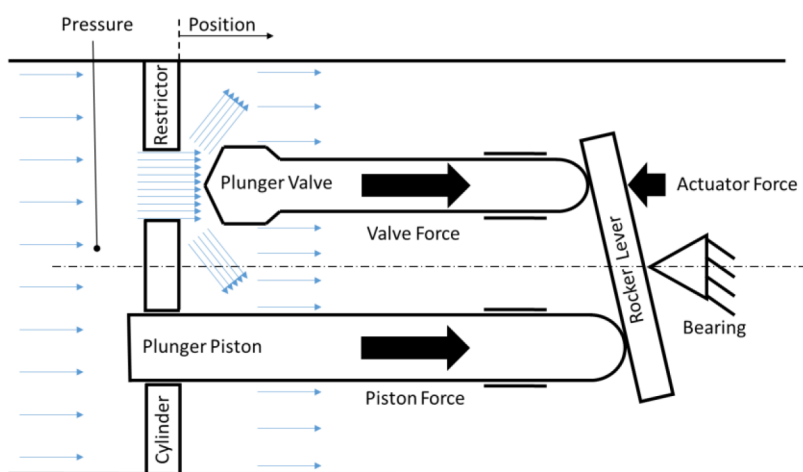


Figure 4—Concept of a force balanced valve

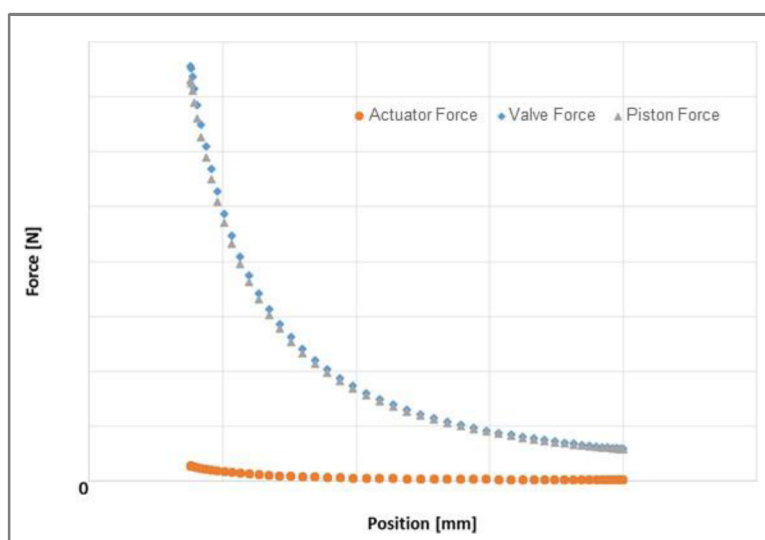


Figure 5—Valve Force, Piston Force, Actuator Force

With valve forces being balanced, the required power to drive such valve is significantly reduced. The actuator drive would primarily have to account for the inertia loads and mechanical losses, such as friction. Figure 6 shows a 3D model of the initial force balanced pulser used for lab testing and prototype field

trials while [Figure 7](#) shows a fully assembled prototype pulser. The valve section made visible through the opening in the 3D model is located in the center of the prototype photograph below.

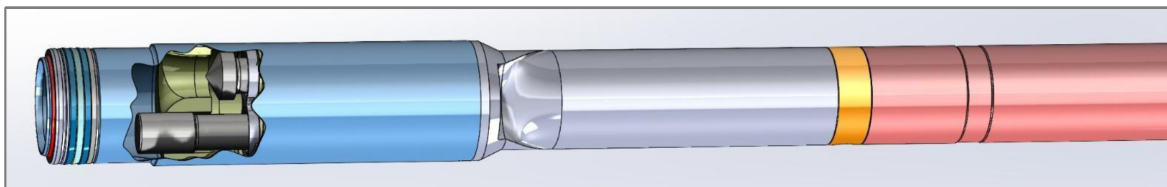


Figure 6—3D model of a force balanced pulser

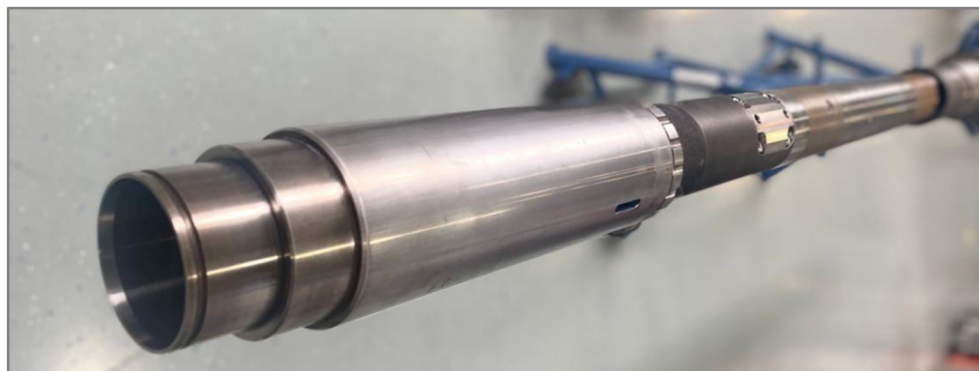


Figure 7—Prototype of force balanced pulser

Signal Modulation

The force balanced valve concept enables highly dynamic continuous valve piston position control at power levels suitable for downhole applications. This allows for a lot more flexibility in the design of the signal modulation scheme compared to systems with oscillating disk valves. The novel valve concept allows for the optimization of the pressure signal shape that represents the bit stream, which increases spectral efficiency and robustness against interfering noise to enable higher physical data rates. Such noise may originate, for example, from the mud pumps.

The main design constraint to be adhered to during the design of the modulation scheme is the preservation of a continuous phase characteristic of the modulated signal. As illustrated in [Figure 8 \(b\)](#), applying a conventional binary phase-shift keying (BPSK) approach to the bit stream shown of [Figure 8 \(a\)](#) results in abrupt phase transitions at bit boundaries. These discontinuities translate into an instantaneous change of the plunger position, which is mechanically impossible to reproduce by a mud pulse valve. Consequently, a continuous phase modulation scheme (CPM) is mandatory for use in mud pulse telemetry. An example modulation scheme for an oscillating disk valve is depicted in [Figure 8 \(c\)](#), addressing the continuous phase constraint by introducing transient symbols at bit boundaries to smoothen the waveform. However, this comes at the cost of reduced spectral and temporal efficiency. With the continuous position control of the force balanced pulser novel modulation schemes can be designed without dedicated signal transitions but with continuous phase adjustment. An example of such a modulated signal is shown in [Figure 8 \(d\)](#). The modulated signal represents the same bit stream of [Figure 8 \(a\)](#), however, with a significantly more gradual phase adjustment than the scheme of [Figure 8 \(c\)](#).

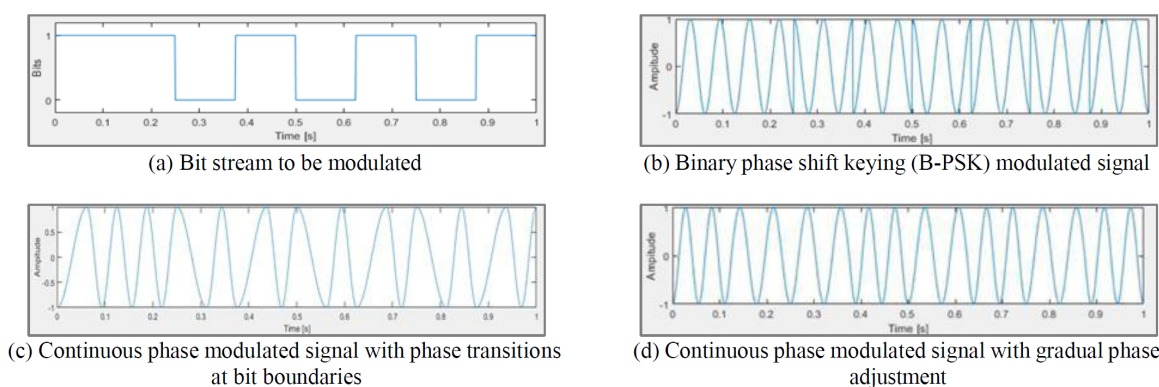


Figure 8—Different modulation schemes representing the bit stream depicted in diagram (a)

In addition to the critical design constraint of phase continuity, the modulation scheme can be optimized for spectral efficiency within the actuation power limits of the mud pulser. This is where the force balancing concept delivers advantages over other valve concepts, since it requires low actuation power for high frequency signaling. As shown in Figure 9 (a) and (b), the power spectral densities (PSD) of the schemes depicted in Figure 8 (c) and (d) differ significantly. The power spectrum of signal with gradual phase adjustment exhibits an ~ 5 Hz narrower main lobe than the signal with phase transitions at bit boundaries. Since the main lobe is used for decoding this narrower bandwidth makes the signal less susceptible to interfering noise, especially between 35 and 40 Hz. Also, the signal with gradual phase adjustments shows reduced side-lobe energy dispersion compared to the signal with phase transitions at bit boundaries. This means that more of the mechanical power invested into signal generation is injected into the main lobe and reaches the surface mud pulse decoder. Both effects result in a higher signal to noise ratio enabling a higher physical data rate.

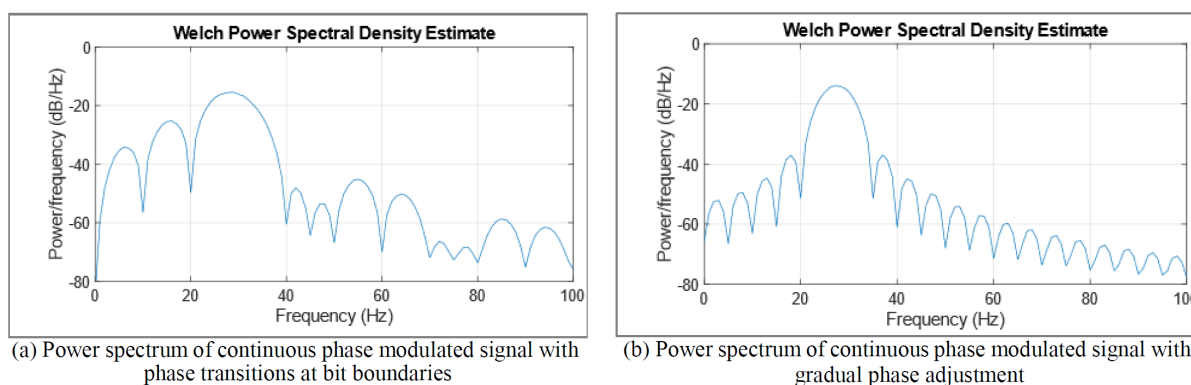


Figure 9—Power spectral density plots for the signaling schemes depicted in Figure 8

Operationally an important additional advantage of the gradual phase adjustment modulation scheme is its high frequency agility relative to the phase transitions scheme. It allows the modulated signal to be centered around any desired carrier frequency. This facilitates signal placement within a clean region of the frequency spectrum, away from interfering pump signal harmonics and signal frequencies that are strongly affected by attenuation. This gives MWD field engineers more opportunity to optimize system configuration and increase the data rate compared to established systems. Figure 10 shows a spectrogram (signal frequency spectrum over time) of a received mud pulse telemetry signal after noise cancellation. Dark blue indicates frequencies with low power, green represents medium power and red high power. The discrete horizontal lines e.g. at 5 Hz and 15 Hz are interfering signals originating from the pumps, which could not be removed completely by pump signal cancellation. The cloudy area between ~ 20 Hz and ~ 30 Hz is the telemetry signal

with a carrier frequency of 26 Hz. With the ability to select an arbitrary carrier frequency, the signal could be placed in a frequency band free of remaining pump interferers, thereby enabling reliable data transmission.

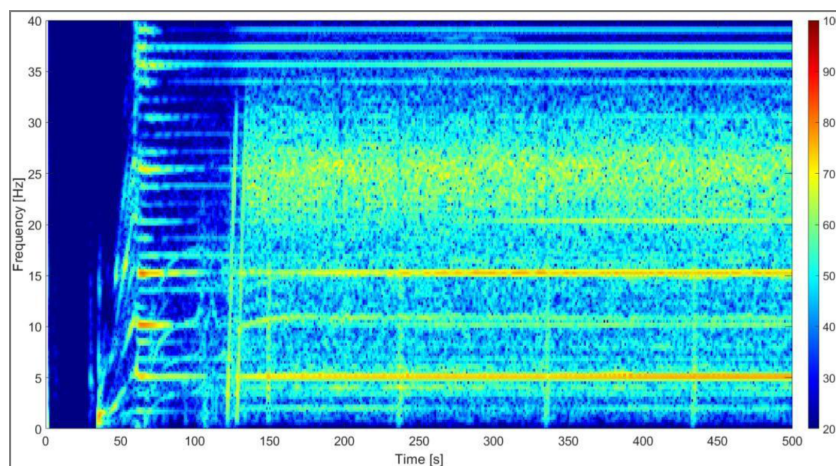


Figure 10—Spectrogram of mud pulse receive signal after noise cancellation

System Verification and Validation

During system design interdependencies between system components were considered as component-level requirements. Also, where feasible, models were used to simulate the interplay of system components. That way the components could be designed and optimized in parallel. Results from component lab tests could be used to tune system models and / or adjust requirements for other system components. In this section we describe some aspects of the conducted lab tests, and we provide a summary of system level testing on a test rig as well as in customer wells.

Lab testing of valve system

The components of the valve system have been designed in CAD and analyzed by Computational Fluid Dynamics (CFD) simulations (Figure 11) in numerous design iterations. Figure 11 shows the fluid velocity of three different design variants of the pulser valve. The fluid velocity and corresponding streamlines are important in understanding and adjusting force reaction, local pressure, turbulence and potential areas of erosion. Promising candidates from the CFD studies were manufactured and tested at the in-house flow loop, verifying the anticipated characteristics and advantages with respect to force balancing performance, pressure build-up and drive power demands.

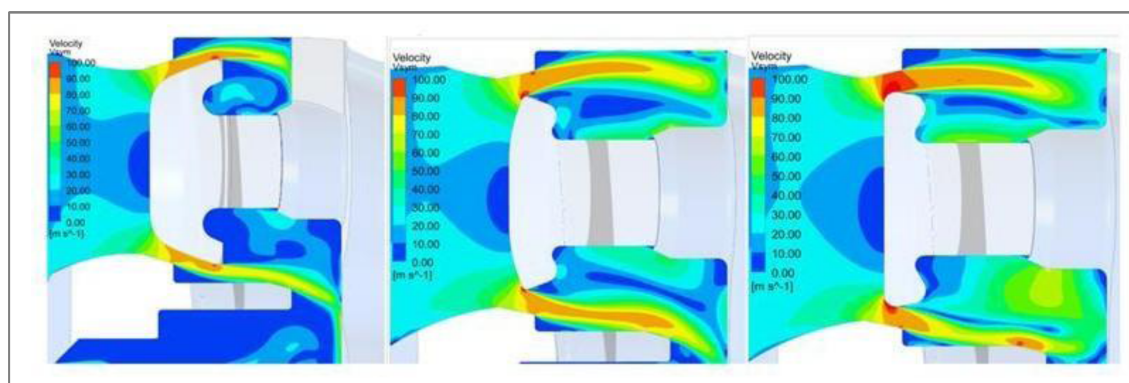


Figure 11—CFD simulation of valve design options

LCM tests. Besides the applicability of the reciprocating valve for high-speed telemetry one major operational advantage of the force balanced valve concept is the expected superiority with regards to solids handling and anti-plugging performance (LCM tolerance). During the iterative valve design process, a series of LCM tests was conducted using different LCM types and concentrations. In these tests various combinations of flow rates and valve positions, and corresponding signal pressures were tested, accompanied by constant monitoring and logging of representative diagnostic parameters for post-test analysis. Different variants of LCM were tested spanning from granular to fibrous materials. Typical test LCM types for qualification include Nut Plug coarse with particle size of up to 8 mm and Cedar Fiber with fiber length larger than 10 mm (Figure 12). It could be shown that the new force balanced pulser can handle extreme concentrations of LCM. Test were stopped at concentrations of 80 ppb, having reached the limits of the flow loop mud circulation system rather than the limits of the pulser system.

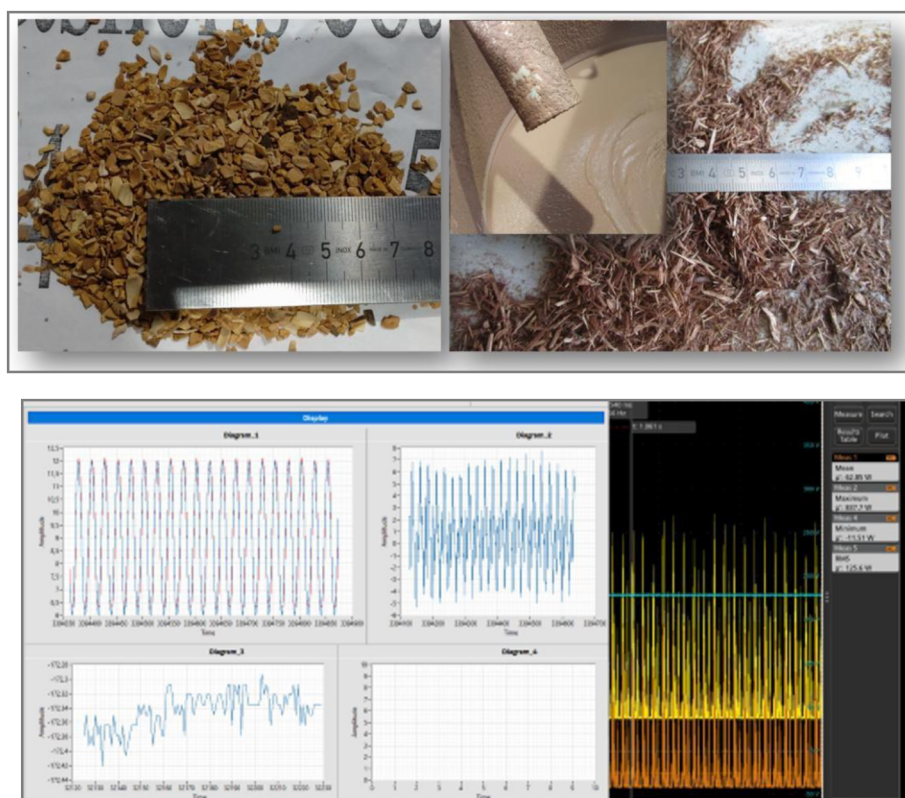


Figure 12—LCM Test at Flowloop, LCM material variants (upper part), Data acquisition (lower part)

High temperature capability & durability verification. The valve assembly and corresponding actuation mechanics were tested in the laboratory to extreme conditions, such as combined temperature and pressure testing in an autoclave (Figure 13) as well as combined temperature and vibration testing on a linear shaker (Figure 14). The autoclave test was setup to not only verify the assembly lifetime by prolonged exposure to extreme conditions, but also to mimic combinations of pressure and temperature that may occur in operating scenarios, such as low temperature and high pressure or high temperature and low pressure. The test program cycled through various scenarios including temperatures of up to 175°C to simulate repetitive use of the tool in its operational life. The linear shaker was used to verify vibration resistance at extreme conditions and combinations. Vibration profiles and acceleration levels used for testing were derived from actual field data, accompanied by a temperature profile to test a temperate range between room temperature and 150°C, in combination with vibration. Tests were conducted at three axes of vibration.



Figure 13—Installation of the force balanced pulser in the autoclave for reliability testing (left: autoclave open, right: autoclave closed)



Figure 14—Installation of the force balanced pulser on a linear shaker for reliability testing

Deployments at Test Rig

After extensive component level lab testing and system integration testing in the flow loop, a couple of drilling trials were conducted at a dedicated full-scale land drilling rig in northern Germany, which is operated for testing of drilling equipment. Goal of the tests was verification of the overall system functionality, first while circulating and later also while drilling, to de-risk follow-up deployments in commercial wells. The test site possesses multiple slots of varying diameters for testing drilling tools of all BHA sizes. The boreholes are planned with a maximum vertical depth of 3,000 m and a maximum horizontal radius of 2,000 m.

The focus for this phase of the system verification was the implementation of the high-speed signal modulation scheme in the pulser, along with its associated decoding algorithms and MWD surface software. The testing campaign comprised:

- Drilling of 7 curve/lateral sections,
- Accumulation of over 500 circulating hours,
- Extensive testing across a wide range of physical data rate and carrier frequency combinations,
- Adjustment of physical data rate and / or carrier frequency via mud pulse downlink,
- Operation under varying drilling conditions, including changes in surface RPM, weight on bit and pump flow rates.

During these tests physical data rates of up to 30 bit/s were achieved in a measured depth of up to 1,711m using a water-based drilling mud. These trials confirmed the robustness and adaptability of the telemetry system under realistic and dynamic drilling conditions.

Testing in commercial wells

Following the functional system verification at the test rig, the next-generation high-speed mud pulse telemetry system was deployed in commercial wells onshore North America. To validate its robustness and operational reliability it was used for performance drilling operations with slim hole rotary steerable BHAs. The system was successfully run in more than 25 wells without pulser failure/trip for failure of telemetry service and with data rates of up to 18 bit/s depending on operating conditions. Tests encompassed both curve and lateral sections, giving exposure to a wide range of downhole conditions:

- Mud weights ranging from 8.0 to 12.5 ppg, with both water-based and oil-based muds,
- Total drilled footage exceeding 250,000 feet,
- More than 1,700 circulating hours,
- Physical data rates reaching 18 bit/s, significantly surpassing the established norm of ~3 bit/s in these wells,
- Multiple "mile-a-day" footages,
- Maximum single-run footage of 18,016 feet,
- Four wells drilled with measured depths exceeding 25,000 feet, with a maximum TD of 27,223 feet,
- Maximum recorded downhole temperature of 124°C.

These deployments confirm operation applicability of the innovative force balanced valve pulser combined with a novel signal modulation scheme and corresponding decoding software. In applications with advanced LWD measurements the bandwidth gains relative to established technologies will allow for increased ROP without compromising log quality.

Summary, Conclusion and Outlook

In this paper we presented a novel mud pulse telemetry system based on the concept of a force balanced mud pulser and continuous phase modulation with gradual phase adjustment. The system addresses the need for a highly reliable high speed mud pulse telemetry system incl. LCM tolerance and high temperature resilience. We showed how a force balanced mud pulser and an optimized signal modulation scheme address these needs. Extensive lab testing at temperatures up to 175°C, LCM testing until reaching the limits of the flow loop test stand as well as deployments in more than 25 commercial wells without tool failures confirmed reliability and capability of the chosen design.

In conclusion the presented system combines the best of two worlds, the reliability benefits of a directly controlled plunger valve with reciprocating plunger with the high-speed capabilities of rotating / oscillating disk valves. This operational versatility marks a significant advancement over conventional mud pulse telemetry systems. Going forward the developed system will be tested in various regions across the globe to operationally verify its high-temperature resilience and integrate it with advanced LWD BHAs to enable elevated ROPs without sacrificing LWD log density.

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